Firmware Analysis Report

Project: firmware_test.bin
Architecture: Unknown
File Size: 4504 bytes
SHA256: 6878fc100602672da53cd3d0b8f03ad448f826bfddb13fa3f6f85bccbc75cafd

Executive Summary

This report was automatically generated by the FAWS system. It contains the results of a multi-stage static, dynamic, and symbolic firmware analysis.

Analysis Stages

Tool: STRINGS

Status: SUCCESS | Command: "C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\.venv\Scripts\python.exe" tools/strings.py --target "C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\data\uploads\26_firmware_test.bin" --project "26" --run-id 0

```
Starting Strings Analysis on C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow
simulator\backend\data\uploads\26_firmware_test.bin... Extracted 114 strings. Found 23
finding(s). [!] IPv4_ADDRESS: OBIS_CODE:1.0.1.8.0.255 [!] PRIVATE_KEY_MARKER: -----BEGIN RSA
PRIVATE KEY----- [!] FILE_PATH: MQTT_BROKER_URL=mqtt://broker.hivemq.com:1883 [!] FILE_PATH:
MQTT_BROKER_TLS_URL=mqtts://iot.eclipse.org:8883 [!] IPv4_ADDRESS:
OTA_UPDATE_URL=http://192.168.1.1/update.php [!] URL:
OTA_UPDATE_URL=http://192.168.1.1/update.php [!] FILE_PATH:
OTA_UPDATE_URL=http://192.168.1.1/update.php [!] URL:
CLOUD_API_URL=https://api.firmware-update.example.com/v1/ota [!] FILE_PATH:
CLOUD_API_URL=https://api.firmware-update.example.com/v1/ota [!] IPv4_ADDRESS: 192.168.1.1 Scan
complete.
```

Tool: BINWALK

Status: SUCCESS | Command: "C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\.venv\Scripts\python.exe" tools/binwalk.py --target "C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\data\uploads\26_firmware_test.bin" --project "26" --run-id 0

```
Starting Python Binwalk Engine on C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow
simulator\backend\data\uploads\26_firmware_test.bin... Found 3 finding(s). [!] Offset 0x0: ELF
32-bit LSB executable [!] Offset 0x9c7: Squashfs filesystem [!] Offset 0x9e6: gzip compressed
data Scan complete.
```

Tool: CUTTER

Status: SUCCESS | Command: "C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\.venv\Scripts\python.exe" tools/cutter.py --target "C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\data\uploads\26_firmware_test.bin" --project "26" --run-id 0

```
Starting Cutter (Lightweight Fallback) on C:\Users\akars\OneDrive\Desktop\Firmware Analysis
workflow simulator\backend\data\uploads\26_firmware_test.bin... Found 1 finding(s). [!]
Architecture: ELF executable detected Scan complete.
```

Tool: GHIDRA

Status: SUCCESS | Command: "C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\.venv\Scripts\python.exe" tools/ghidra.py --target "C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\data\uploads\26_firmware_test.bin" --project "26" --run-id 0

```
Starting Ghidra (Lightweight Fallback) on C:\Users\akars\OneDrive\Desktop\Firmware Analysis
workflow simulator\backend\data\uploads\26_firmware_test.bin... No interesting RE artifacts
found. Scan complete.
```

Tool: TRUFFLEHOG

Status: SUCCESS | Command: "C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\.venv\Scripts\python.exe" tools/trufflehog.py --target "C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\data\uploads\26_firmware_test.bin" --project "26" --run-id 0

```
Starting Secret Scan on C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow
simulator\backend\data\uploads\26_firmware_test.bin... Found 5 finding(s). [!] GENERIC_SECRET
at line 3: TOKEN=ghp_FAKEtoken1234567890abcdefFAKE [!] GENERIC_SECRET at line 3:
password='Sup3rSecretPass [!] GENERIC_SECRET at line 3: PASSWORD=FactoryReset [!]
GENERIC_SECRET at line 13: TOKEN=eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJzdWIiOiJkZXZpY2UtMDAx
Iiwicm9sZSI6ImFkbWluIiwiaWF0IjoxNzAwMDAwMDAwfQ.RkFLRVNJR05BVFVSRUJZVEVTTk9UUkVBTA [!]
GENERIC_SECRET at line 13: PASSWORD=WifiPass2024 Scan complete.
```

Tool: ENTROPY

Status: SUCCESS | Command: "C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\.venv\Scripts\python.exe" tools/entropy.py --target "C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\data\uploads\26_firmware_test.bin" --project "26" --run-id 0

```
Starting Entropy Analysis on C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow
simulator\backend\data\uploads\26_firmware_test.bin... Overall Shannon Entropy: 6.2502 Entropy
is within normal range for executable code. Scan complete.
```

Tool: WIRESHARK

Status: SUCCESS | Command: "C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\.venv\Scripts\python.exe" tools/wireshark.py --target "C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\data\uploads\26_firmware_test.bin" --project "26" --run-id 0

```
Starting Wireshark (Lightweight Fallback) on C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\data\uploads\26_firmware_test.bin... No interesting network packets found. Scan complete.
```

Tool: AFL++

Status: SUCCESS | Command: "C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\.venv\Scripts\python.exe" tools/afl.py --target "C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\data\uploads\26_firmware_test.bin" --project "26" --run-id 0

```
Starting AFL++ (Lightweight Fallback) on C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\data\uploads\26_firmware_test.bin... No obvious fuzzing entry points found. Scan complete.
```

Tool: ANGR

Status: SUCCESS | Command: "C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\.venv\Scripts\python.exe" tools/angr.py --target "C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\data\uploads\26_firmware_test.bin" --project "26" --run-id 0

```
Starting angr (Lightweight Fallback) on C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\data\uploads\26_firmware_test.bin... Found 1 finding(s). [!] Symbolic: High number of branch instructions (144) indicates complex control flow suitable for symbolic execution Scan complete.
```

Tool: SCORECARD

Status: SUCCESS | Command: "C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\.venv\Scripts\python.exe" tools/scorecard.py --target "C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\data\uploads\26_firmware_test.bin" --project "26" --run-id 0

```
Starting Risk Scorecard Generation on project 26... Found 1 finding(s). [!] RiskScore: Calculated Aggregate Risk Score: 0/100 based on 20 findings Scan complete.
```

Tool: PDF_REPORT

Status: RUNNING | Command: "C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\.venv\Scripts\python.exe" tools/pdf_report.py --target "C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\data\uploads\26_firmware_test.bin" --project "26" --run-id 0